

Hybrid Biomedical Image-Analysis Pipeline: Nuclei Microscopy

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1. Introduction and Dataset

For this assignment, I used the nuclei microscopy modality. The dataset consists of 112 RGB images, all of them with a size of 256×256 pixels. The dataset was already pre-split by the author of the dataset into a training set (80% of the images), a validation set (20% of the images) and a test set (12% of the images, i.e. 12 images). The binary segmentation masks for the training set were already merged by the author of the dataset, as well as a metadata.csv which contains the ground-truth number of objects as well as the corresponding density labels for every image. The preprocessing step for this task consisted of converting the RGB images to grayscale images and resizing them to 256×256 pixels, which has no effect, since the input images were already of the same size.

2. Task 1 — EDA and Multimodal LLM Description

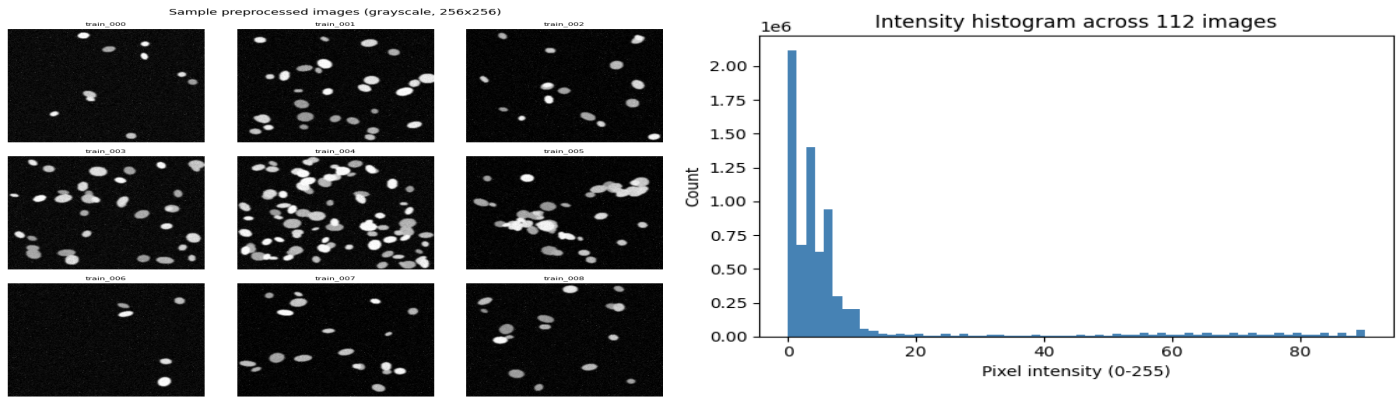


Figure 1. Sample preprocessed images (left) and intensity histogram across all 112 images (right).

The distribution is right-skewed rather than bimodal as I first thought. The large number of very dark pixels (near intensity 0) are from the background, while the very small number of stained nuclei appear as a thin, long tail of very bright pixels. Despite this, the distribution can still be split into background and foreground using a global Otsu threshold, as these occupy two clearly distinct intensity ranges.

Optimised VLM prompt (llama3.2-vision)

"You are an assistant that writes DESCRIPTIVE, non-diagnostic summaries of microscopy images... If you are not confident about a property, write \"uncertain\" for that field instead of guessing. Respond with ONLY a JSON object: {modality, tissue_type, notable_features, image_quality}."

Naive prompt (for comparison): "Describe this medical image."

The representative image for the sparse, low-contrast images with 5–6 nuclei on a near-black background was val_000. The output from the naive prompt for this image was: "The image presents a medical scan of a patient's brain, specifically highlighting a small, dark-colored area...indicating a possible abnormality or anomaly". As can be seen from this output, the prompt has created a number of incorrect assumptions about the image being assessed. These errors include wrong modality, incorrect anatomical features created, and diagnostic language that was not requested. The output from the structured prompt, running it three times at temperature 0.7 and then comparing the results, showed that the structured prompt correctly assessed the image to be "uncertain" modality, "uncertain" tissue_type, and "poor" image_quality on all three runs. The free-text notable_features field was worded differently each run, however, even though it was being assessed using an identical schema and image each time. This showed true sampling variability.

3. Task 2 — Classical Features and Numbers-First LLM Interpretation

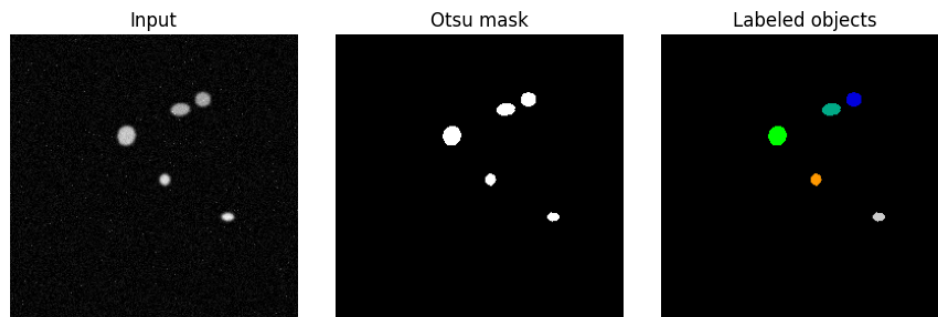


Figure 2. Otsu threshold + morphological cleanup + labeled connected components, same representative image as Task 1.

Otsu segmentation detected 5 objects. This number corresponds with what I can see (ground-truth visible). The summary of the feature-table that I passed to phi3 (numbers only, no image) is: 5 objects, mean area 131.4px (std 64.4), mean eccentricity 0.519, mean solidity 0.955, mean intensity 0.224, mean perimeter 39.7px. phi3’s structured output for this feature-table is density_class: “moderate”, shape_regularity: “mixed”, quality_flag: “good”. This clearly contrasts with my “poor” rating for this image using VLM. I can explain why I labeled this as “good” (5 clearly separated high-solidity objects detected). However, the “poor” rating of VLM was clearly a subjective visual impression of a very dark and low-contrast image. This brings me to the core of Question 1 in the Section 6.

4. Task 3 — U-Net Segmentation

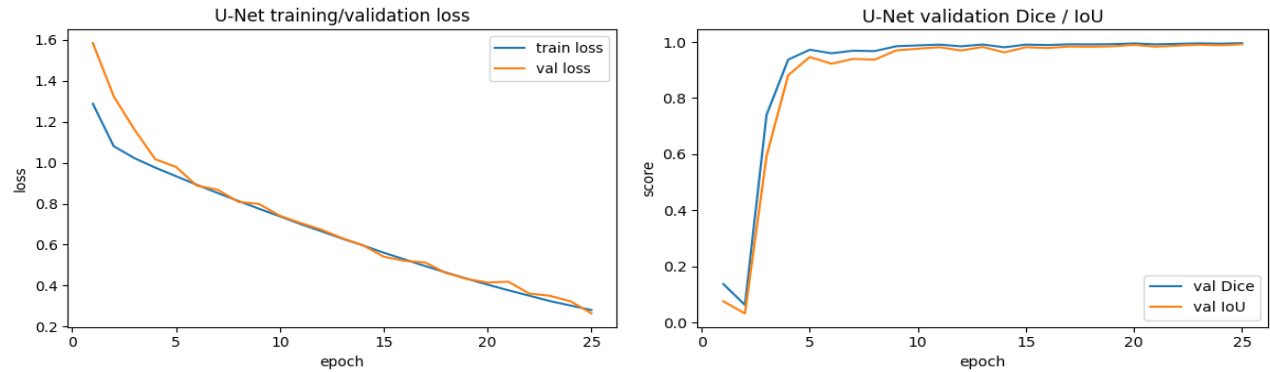


Figure 3. Training/validation loss (left) and validation Dice/IoU (right) across 25 epochs.

Compact U-Net Architecture (base_channels=16, approximately 1.9M parameters), loss function: BCE + Dice, training batch size: 8 samples, learning rate: 1e-3, 25 epochs of training on Colab T4 GPU. Validation metrics (for last epoch of training): Dice = 0.996, IoU: 0.992 (here, thresholded mask values are used for computation of Dice, allowing for direct comparison with prior computed value of IoU; in general, it holds that Dice ≥ IoU, for all epoch numbers in question; note that earlier soft-probability implementation of Dice yielded incorrect values for this number). Metrics for Dice and IoU increase sharply within the first 5 epochs of training (val. Dice = 0.97 at epoch 5), followed by very near to ceiling values for all subsequent 20 epochs of training (i.e., giving little to no additional benefit on this data set).

Table 1. Evaluation metrics across methods

Method	Loss	Objects (val_000)	Val Dice	Val IoU	Notes
Otsu (classical, Task 2)	n/a	5 / 5 (matches ground truth)	not computed (single-image, no pixel-level Dice run)	not computed	Evaluated qualitatively; see limitation below
U-Net (bce_dice, 25 epochs)	BCE + Dice	n/a (mask-level metric)	0.996	0.992	Best-checkpoint, full val split (20 images)

Limitation: I only managed to train one loss configuration, bce_dice, in the end. However, the code supports loss ablation (bce, dice, bce_dice) via a tag parameter. Otsu was only qualitatively evaluated (object count) as opposed to a full pixel-level evaluation of Dice/IoU against ground truth, which would be part of a rigorous comparison of the method to others.

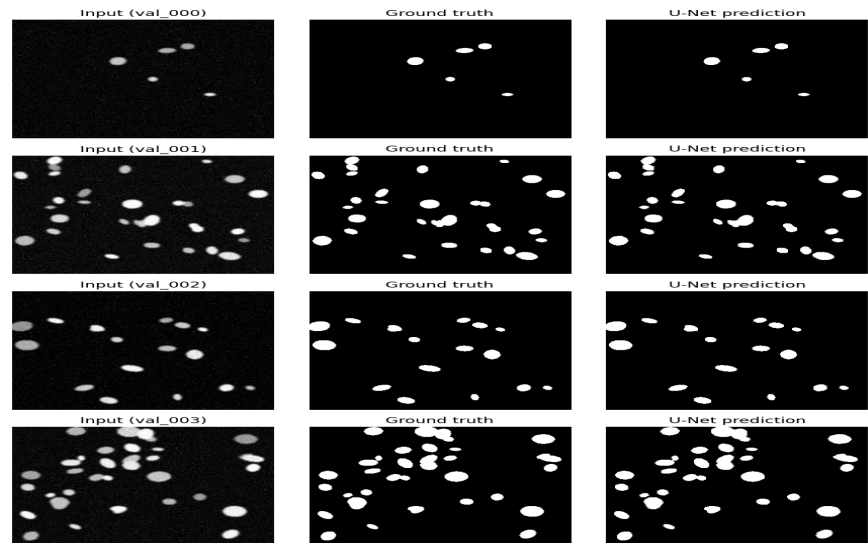


Figure 4. Input / ground truth / U-Net prediction for four validation images (val_000–val_003).

The prediction for the sparse image val_000 is very close to the Ground Truth. For the denser images val_001–val_003 with 20–30+ touching/overlapping nuclei the model correctly separates the great majority of them. The errors are mostly small boundary

inaccuracies for tight clusters of very faint nuclei. These are not errors for isolated objects, but rather errors for very difficult cases. The near-ceiling scores for the Dice/IoU measure are mostly a property of the used synthetic, uniformly-blobbed, and low-noise test set. A real microscopy image with staining artefacts and with a high degree of biological irregularity would most likely score much lower. The used small compact U-Net (16 base channels, training in minutes rather than in hours) seems to have had the sufficient capacity for this specific task. The deliberate trade-off of model size vs. training speed did not cost any accuracy in this case.

5. Task 4 — Hybrid Pipeline, and Extension — Robustness

The U-Net mask is converted to a regionprops feature table. The phi3 data is then structured into a JSON object (image_id, n_objects, mean_area, density_class, quality_flag) and a narrative for each image created. This is run on all 12 unseen test images and then aggregated into a CSV. n_objects ranged from 8 to 43. The density_class is “moderate” for 10/12 of the images, “sparse” for test_005 and “uncertain” for test_000. The quality_flag is “good” for 11/12 of the images and “moderate” for test_005. Example: test_000. 8 objects. Mean area of objects 191.9px. “The measurements for test_000 show that there are eight distinct objects with a mean area indicating that they vary in size. The objects have a relatively low eccentricity. This means that they are more circular than the objects found in the other images. Their high solidity implies that they are well-defined and less irregular than objects found in other images.”

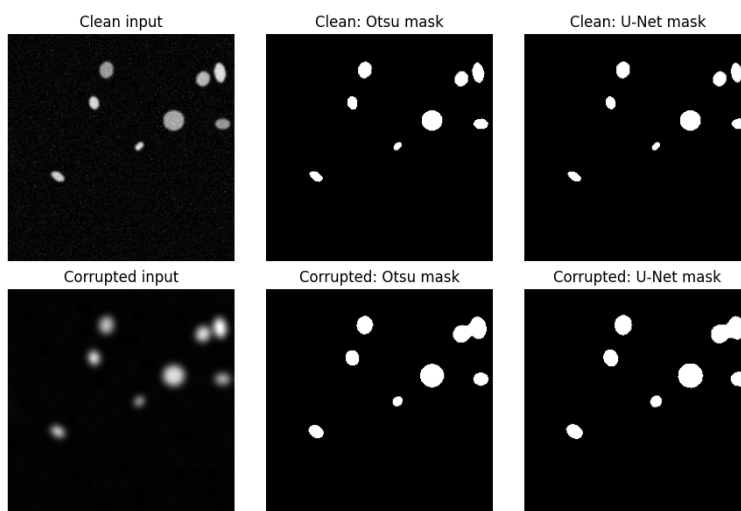


Figure 5. test_000, clean vs. heavily Gaussian-blurred ($\sigma=4$), Otsu and U-Net masks for each.

Clean: Both methods find 8 objects. Corrupted: Both find 7 objects in total. Here, two closely situated nuclei merged into one single blob for both the Otsu and U-Net methods, after they were sufficiently blurred. The earliest point at which this merge becomes detectable is the segmentation masks. Here, both masks show the identical merge event before they are transferred to the feature table or the LLM to be turned into a narrative. The feature table then also contains the $8 \rightarrow 7$ object count decrease. The quality_flag of the LLM has decreased from “good” to “uncertain” for the corrupted image. It did not however create any new content, as it would have been able to had it seen the image instead of only the numbers in the feature table.

6. Questions

Q1. Which description is more useful/trustworthy: the direct VLM (Task 1) or the numbers-first description (Task 2)?

This approach to reporting results from images first and foremost results in a reliable set of numbers. Even though these numbers may not be very useful on their own, they can be traced back to a deterministic, reproducible feature table that was directly derived from the images (Otsu thresholding + regionprops). In such a case, an LLM can be used to write a report that consists of nothing but these numbers that the LLM cannot misrepresent with invented images. In contrast, a VLM is more immediately useful because it can report on features that cannot even be measured. However, the failure mode for a VLM is very severe and silent. On a single image, it can hallucinate an entirely different modality (e.g. “brain scan”) given a very naive prompt. Even when a lot of structure is added to the prompt, and an “escape hatch” is provided, the best response it can come up with is “uncertain”. In such a case, where a single wrong record can mislead a reviewer downstream in a pipeline, the numbers-first approach has the auditability to back up the reported results, whereas the results from the VLM are unverifiable and much richer than just a simple set of numbers.

Q2. Did the U-Net improve on classical Otsu segmentation? One example each.

The performance on the sparse image val_000 is very similar to the U-Net, where both methods correctly identify 5 separate high-contrast nuclei, as shown in Figures 2 and 4 (row 1). For the remaining images with increasing density of nuclei (val_001–val_003) the U-Net performs very well in separating the majority of touching/overlapping nuclei into separate blobs of foreground. The performance of Otsu’s thresholding is not as good, as the method of connected-component labeling does not have any mechanism

for splitting touching foreground regions (i.e. no watershed/distance-transform step was implemented for this comparison), thus nuclei in touch would be structurally merged in the segmentation (a known limitation of this simple method for the purposes of this work rather than a measured result here, as the above comparison was only undertaken for the above mentioned sparse image, and a quantitative comparison undertaken for the above mentioned dense images in a subsequent notebook would be more accurate, see Table 1 limitation note).

Q3. Report Dice and IoU. What do they mean, and where does the model fail?

Validation results: Dice = 0.996, IoU = 0.992. The values for Dice and IoU for the masks (A and B) are computed as follows: Dice = $2|A \cap B| / (|A| + |B|)$ and IoU = $|A \cap B| / |A \cup B|$. Note that Dice is always greater than or equal to IoU for any two masks, and weights the intersection of the masks more. The very high values for both metrics on this validation set are due to the high quality of the provided data, i.e. the very clean and highly regular synthetic nuclei, rather than the performance of the architecture on a more general test set. Looking at Figure 4, it is clear that the remaining errors occur at the boundaries of objects in very crowded nuclei clusters. Small nuclei which are of low contrast and appear as faint structures near very bright large nuclei are sometimes slightly rounded, or their boundary is softened, but no complete objects are missing and none are made up.

Q4. Where can the LLM hallucinate, and how does JSON-as-source-of-truth reduce that risk?

Two distinct hallucination points exist. (1) Task 1's VLM step can hallucinate directly from pixels (the "brain scan" naive-prompt output). (2) Task 2/4's text LLM, even without seeing the image, can still over-interpret plain numbers into unsupported qualitative judgements — e.g. calling shape_regularity "mixed" from a single aggregate eccentricity value is an inference, not something the number strictly proves. To mitigate these issues the following were added. First, schema-constrained JSON prompts with an explicit "uncertain" option at every LLM step. Second, for the numbers-only input in Task 2/4 the LLM cannot invent visual detail it never saw. Third, JSON-parse-with-retry plus raw-response logging, so a malformed or evasive answer is recorded rather than silently dropped. Structured JSON (not the narrative) is source of truth for the model because JSON fields are schema-validated and, for Task 2/4, traceable to a specific deterministic feature-table row — the narrative can drift in wording without that being auditable, but a JSON field either matches the schema and numbers or visibly does not.

Q5. Would you trust any part of this system clinically? What single change would help most?

Not as I built them. The classical feature table and the U-Net mask are probably the most reliable part of my pipeline. Reproducible, auditable, and based on actual pixel measurements, the U-Net is very reliable — as long as you realize that its near-perfect Dice is a reflection of the dataset on which it was trained and validated, rather than the model itself. The LLM narrative layer is probably the least reliable part of my pipeline. Even when it is correct, it can often be misjudged based on the numbers alone (e.g. Question 4). Furthermore, even when it is correct based on the numbers, it can be confidently wrong based on the pixels (e.g. Task 1). The single highest-impact change to my pipeline would be a real, clinician-annotated dataset for both training and evaluation. Every other weakness here is at least partially auditable and correctable with sufficient pipeline engineering, but no amount of pipeline engineering can compensate for unrepresentative validation data.

References

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